prompting-explained

Please, Thank You, and Other Words the AI Ignored

A teaching tool that shows you the impact of each phrase in your prompt on the AI's answer.

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github.com/Vihaan8/prompting-explained

01 Executive summary

AI tools are now used well beyond data and AI teams, and most of their users can't see how the model interpreted their prompt.

prompting-explained shows people which phrases in their own prompt changed the answer. It removes one phrase at a time, reruns the prompt, scores each answer, and only reports effects larger than the AI's normal run-to-run variation.

Across 8 everyday tasks, prompt wording alone moved the AI's average score from 0.59 (novice prompts) to 0.97 (expert). Nearly half of all phrases asked for things the AI did anyway; format and task instructions mattered most, politeness and tone barely at all. A deletion test confirmed the tool points at the phrases that really matter.

02 The problem

Users can't see how the AI read their prompt. Teachers, marketers, nurses, students and support agents now use AI every day, but the tools give no feedback on which words shaped the answer.

Common advice doesn't hold up. In our tests, "please" and "thanks" almost never changed the answer, and "dont guess" made the AI speculate *more*. Without feedback, people can't tell which of their habits help and which backfire.

Mistakes reach real work. Asked to "turn this ticket into a bug report" for a public tracker, the AI included the customer's name in 2 of 3 runs, and their email in one. Nothing in the answer warned the user.

03 Who it's for

Anyone learning to prompt. Practice on everyday tasks and see how the AI read your words.

Companies. Add your own tasks, documents and rules, so staff learn on the work they actually do.

Governance teams. Test whether a safety instruction works before rolling it out.

GOAL

Help anyone see how the AI read their words, so they get better at prompting.

RESEARCH QUESTION

Which parts of an everyday prompt actually change an AI's answer, and can removing parts and rerunning explain that **faithfully** and **stably**?

REFERENCES

Zeiler & Fergus (2014), occlusion · Ribeiro et al. (2016), LIME · Lundberg & Lee (2017), SHAP · Petsiuk et al. (2018), deletion test · DeYoung et al. (2020), faithfulness · Alvarez-Melis & Jaakkola (2018), stability · Turpin et al. (2023), unfaithful model explanations

04 Methods

How the tool works out what each phrase did, in four steps. Starred terms are explained in the purple notes.

1 Split the prompt into phrases

An AI splits the prompt into phrases and tags each one (task, audience, format, politeness...). These phrases are the pieces the tool will perturb*. Code keeps each phrase exactly as typed, typos included.

*** Perturbation:** changing part of an input to see how the output responds. LIME does this with single words.

2 Remove one phrase and rerun

The tool deletes one phrase, runs the shorter prompt again, and compares the two answers. This is occlusion*. A phrase's effect is how much the answer's score drops without it, so every phrase gets its own counterfactual**: what if you hadn't written this?

*** Occlusion (leave-one-out):** hide one part of the input, rerun, and measure the change.
**** Counterfactual:** the same case with one thing changed, to see what that one thing did.

3 Score every answer two ways

Against the task. Each task comes with a short checklist that code grades automatically, like a teacher's rubric. Every answer gets a score from 0 (misses everything) to 1 (passes everything).

Against your instructions. Code can't check everything you might ask for, such as "use a candy example". So a second AI reads each answer and ticks off which of your instructions it followed*. That tells "the AI followed this" apart from "this made the answer better".

*** AI as a judge:** using one AI to grade another's output. It can be wrong, so its mistakes are reported.

4 Filter out noise

The same prompt gives a slightly different answer each time, so every version runs more than once. A change only counts if it's bigger than the noise floor*. Anything smaller stays grey, and the tool says "not clear" rather than guessing.

*** Noise floor:** the smallest change the tool can tell apart from the AI's natural randomness (two standard errors).

05 Worked example

Task: write a prompt that gets the AI to explain inflation to your 10-year-old cousin. One user wrote this:

Hi there! Explain inflation to my 10-year-old cousin. Use exactly 3 bullet points. Mention a candy bar in one of them. Use proper technical terms.

Made the answer better · Followed, no change in score · Worked against the task · No real difference

Green · "Explain inflation": without it, the AI explained how the body uses food for energy.

Blue · "Mention a candy bar": the AI did it, but the score was the same either way.

Red · "Use proper technical terms": the AI used words like "Consumer Price Index", too hard for a 10-year-old. Without it, the score rose from 0.40 to 0.78.

Grey · "Hi there!": a greeting gives the AI nothing to act on.

Scores run 0 to 1. On a fresh rerun, 5 of 6 verdicts repeated. The "without it" sentences are written by an AI and labelled as descriptions; every number is computed by code.

06 What we learned about the AI

240 prompts on all 8 tasks, written the way novice, average and expert users would, plus 32 deliberately messy prompts.

1 The wording decided how well the AI did

Same AI, same tasks. Only the prompt changed, and the average score went from 0.59 to 0.97. With no document attached to lean on, it went from 0.36 to 0.94.

0.59	0.77	0.97
novice prompts	average prompts	expert prompts

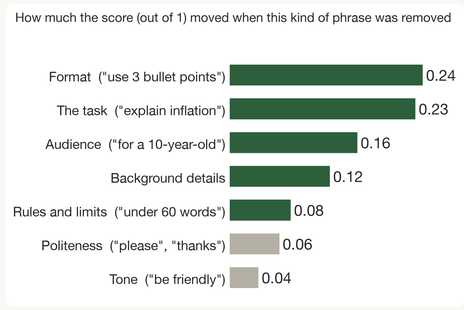
2 Nearly half of what people wrote, the AI did anyway

Of 741 phrases, many asked for things the AI was going to do anyway. Some made the answer worse.

29%	45%	6%
improved the score	the AI did anyway, unasked	made it worse

3 Format and the task mattered; politeness didn't

Removing a format instruction or the task itself moved the score most. Removing politeness or tone barely moved it.



4 The AI didn't always take instructions literally

"dont guess" made it speculate *more*, in all 5 runs. Told to "make up a salary", it refused. "for github issues" made it leave a customer's name out of a bug report without being asked.

Can you trust these findings?

Yes, mostly. Deleting the 2 phrases the tool ranked highest dropped the score by **0.30**, against **0.18** for 2 random phrases (79 prompts, p = 0.0002), so its explanations are faithful*. A full rerun picked the same top phrase **78%** of the time, and 86% when its effect was clear, so they are stable**. Nearly tripling the reruns only lifted agreement from 83% to 91%, so 3 runs is enough.

*** Faithfulness:** the parts an explanation points to really drive the output.
**** Stability:** the same input gets the same explanation when run again.

07 Business implications

Teach by feedback. Show people what their own prompts do, instead of handing out tip sheets.

Teach the right things. Format, the task and the audience first; politeness and tone don't move the result.

Check before trusting. Measure safety instructions, because some backfire.

08 Limitations and next steps

- One model (Claude Haiku 5.5), on tasks with checks written in code.
- Each phrase is compared only with deleting it, not with a better phrase.
- Phrases that do nothing alone are retested as one group, but not in every combination. Shapley values* would, at a much higher cost.
- The splitter and instruction checker are AIs and can be wrong. Checked against code, the checker agreed 88% of the time but can't count words.
- **Next:** Shapley-style scoring, a study with real learners, more models.

*** Shapley values:** give each part a fair share of the credit by testing it in every combination of parts.